

Deliverable 1 — Requirements Table

Customizable Subscription Box Scheduler • Requirements Engineering & UML Use-Case Modelling

Name	SRN	Course	Problem Statement
Prajwal G Devadiga	PES1UG24CS590	Software Engineering - Lab 1	Problem Statement #35 Retail, E-Commerce & Finance

Scenario: A subscription box portal where subscribers customize monthly product selections (e.g. books, coffee, snacks) based on preference tags, with support for pausing or skipping billing cycles.

Req ID	Type	Description (“The system shall ...”)	Priority	Acceptance Criteria (measurable pass/fail)	Rationale	Comments (peer critique → revision)
FR-001	Functional	The system shall allow a subscriber to swap items in the upcoming monthly box or pause that delivery, at any point up to 48 hours before the box's billing renewal timestamp (measured in UTC).	High	PASS: a swap or pause submitted at renewal minus 48:00:01 is accepted, the stored box contents change, and the cycle's fulfillment manifest is refreshed within 5 minutes. FAIL: the request is rejected inside the window, or the manifest still shows the previous contents after 5 minutes.	Late-cycle customization is the core value proposition of the product. 48 hours is the latest point at which the warehouse can still re-pick a box before wave planning.	Peer: "Whose clock is the 48 h measured on?" -> revised to state the renewal timestamp explicitly in UTC. (Given requirement, refined.)
FR-002	Functional	The system shall allow a subscriber to add or remove preference tags (e.g. dark-roast, sci-fi, nut-free) drawn from the curated tag vocabulary, and shall regenerate the recommended contents of the next unlocked box from the updated tag set.	High	PASS: after a tag is saved, the next-box recommendation is regenerated and every recommended SKU satisfies all exclusion tags. FAIL: any recommended SKU violates an exclusion tag, or the recommendation is unchanged after save.	Preference tags are the only input that drives personalization; exclusion tags such as nut-free additionally carry an allergen-safety obligation.	Peer: "nut-free is a safety constraint, not a taste preference." -> exclusion tags are now applied as hard filters, not ranking weights.
FR-003	Functional	The system shall allow a subscriber to skip the next billing cycle up to 3 times in any rolling 12-month period, deferring the renewal date by exactly one cycle without raising a charge.	Medium	PASS: on the 3rd skip the renewal date moves forward one cycle, no payment authorization is sent, and the subscription stays Active; a 4th skip within the same 12 months is refused with reason code SKIP_LIMIT_REACHED. FAIL: a charge is raised for a skipped cycle, or a 4th skip is accepted.	Skipping retains subscribers who would otherwise cancel outright, while the cap keeps recurring revenue forecastable.	Peer: "Table says nothing about the 4th attempt." -> explicit refusal reason code added to the acceptance criteria.
FR-004	Functional	On a box's renewal date the system shall lock the box contents, submit one payment authorization for the subscription amount to the Payment Gateway, and retry a soft decline at most twice at 24-hour intervals before suspending the subscription.	High	PASS: contents are frozen at the renewal timestamp, at most 3 authorizations are attempted, and after the 3rd consecutive decline the subscription moves to Suspended and the subscriber is notified. FAIL: contents change after lock, a 4th authorization is attempted, or a fully declined subscription remains Active.	Recurring billing is the revenue event; bounded retries recover soft declines without breaching card-scheme retry limits.	Peer: "Define soft vs hard decline." -> hard declines (stolen / closed card) are excluded from retry and suspend immediately; recorded in the glossary.
FR-005	Functional	The system shall allow the Fulfillment Lead to generate a fulfillment manifest for a selected billing cycle, grouped by warehouse zone, and export it together with the corresponding shipping labels.	High	PASS: the manifest lists every paid, non-skipped box for the cycle exactly once, grouped by zone, and the exported label count equals the manifest line count. FAIL: a paid box is missing or duplicated, or label count does not equal line count.	The warehouse cannot pick or ship without a per-cycle manifest; it is the hand-off point between the portal and physical fulfillment.	Peer: "Do paused / skipped boxes appear on the manifest?" -> scope narrowed to paid, non-skipped boxes only.
NFR-001	Nonfunctional - Performance & Security	The fulfillment manifest generator shall export the shipping labels for a 10,000-box monthly cycle in under 60 seconds over an authenticated TLS 1.2+ session, while the portal is serving peak subscriber traffic.	High	PASS: under simulated peak load, 10 consecutive benchmark runs complete with a 95th percentile under 60 s, and an unauthenticated call to the export endpoint returns HTTP 401 with no label data. FAIL: the 95th percentile exceeds 60 s, or any unauthenticated request returns label data.	Labels are produced in a narrow window on dispatch day; a slow export delays every shipment, and an unprotected one exposes 10,000 subscriber addresses.	Peer: "Is 60 s an average or a worst case?" -> restated as the 95th percentile over 10 runs. (Given requirement, refined.)
NFR-002	Nonfunctional - Reliability & Availability	The subscriber customization service shall achieve at least 99.9% availability measured per calendar month, and shall not be taken offline for planned maintenance during the 72 hours preceding any billing renewal date.	High	PASS: the monthly uptime report shows 43 minutes or less of unplanned downtime and zero planned maintenance windows inside any 72-hour pre-renewal freeze. FAIL: downtime exceeds 43 minutes, or a maintenance window overlaps a freeze period.	FR-001 gives subscribers only a 48-hour window to change a box; an outage inside that window silently removes a contractual right and is a direct cancellation driver.	Peer: "Tie the freeze window to the 48 h rule in FR-001." -> freeze widened to 72 h so it fully covers the customization cut-off plus a safety margin.

FR-001 and NFR-001 are the requirements supplied with Problem Statement #35; FR-002–FR-005 and NFR-002 are student-authored. The Comments column records the peer-critique round (Lab step 3) and the revision each comment produced.

Actors and Use Cases extracted from the approved requirements

Actor	Classification	Goal in the system
Subscriber	Primary	Customizes the box, manages preference tags, pauses or skips billing cycles.
Fulfillment Lead	Primary	Generates the per-cycle manifest and exports the shipping labels.
Billing Scheduler	Secondary (time actor)	Fires on each renewal date and starts the billing run.
Payment Gateway	Secondary (external system)	Authorizes, declines or refunds the subscription charge.

UC ID	Use Case	Actor / relationship	Traces to
UC-01	Customize Monthly Box	Subscriber	FR-001, FR-002
UC-02	Manage Preference Tags	Subscriber	FR-002
UC-03	Pause or Skip Billing Cycle	Subscriber	FR-001, FR-003
UC-04	Process Renewal Billing	Billing Scheduler	FR-004
UC-05	Generate Fulfillment Manifest	Fulfillment Lead	FR-005, NFR-001
UC-06	Validate Customization Window	(included by UC-01, UC-03)	FR-001, NFR-002
UC-07	Authorize Payment	Payment Gateway	FR-004
UC-08	Export Shipping Labels	(included by UC-05)	FR-005, NFR-001
UC-09	Suggest Substitute Item	(extends UC-01)	FR-002
UC-10	Apply Loyalty Discount	(extends UC-04)	FR-004

Stereotype	From	To	Why
«include»	UC-01	UC-06	every customization must first check the 48 h window
«include»	UC-03	UC-06	pause / skip is bound by the same 48 h window
«include»	UC-04	UC-07	a renewal always authorizes a payment
«include»	UC-05	UC-08	a manifest is always exported with its labels
«extend»	UC-09	UC-01	condition: chosen SKU is below safety stock
«extend»	UC-10	UC-04	condition: 12+ consecutive paid cycles